

Exploratory Computational Research Report

Latent Physiological Topology

Dynamic State-Space Modeling of Cardiac Acoustic Signals

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github.com/MinervaRose/latent-physiological-topology

Abstract

This report presents an exploratory computational framework for modeling cardiac acoustic signals as trajectories through latent physiological state-space rather than as isolated waveform events. Instead of treating phonocardiogram recordings solely as static acoustic classification targets, the framework investigates whether cardiac acoustic activity may exhibit measurable latent organization when represented as dynamical traversal through physiological topology.

Using nonlinear manifold learning, physiological trajectory reconstruction, unsupervised state discovery, probabilistic transition analysis, anomaly detection, toroidal cyclic embedding, orbital phase estimation, and holographic density rendering, the framework explores whether recurrent physiological structure emerges within latent acoustic geometry.

A subset of the CirCor DigiScope Phonocardiogram Dataset was analyzed using computational acoustic segmentation and multidimensional feature extraction. Projection of the resulting feature space into latent physiological topology revealed recurrent state-space regions, directional transition persistence, dynamically unusual anomaly structures, partially cyclic manifold recurrence behavior, and distributed occupation topology.

Importantly, the framework remains exploratory and computational rather than clinically validated. The project does not claim diagnostic capability, pathological inference, or physiologically validated latent topology. Instead, the broader contribution lies in proposing a systems-oriented computational perspective in which physiological acoustic activity may be investigated as dynamically evolving structure embedded within latent geometric state-space.

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1. Introduction

Traditional computational analysis of cardiac acoustic signals frequently emphasizes waveform classification, murmur detection, or diagnostic categorization. Although such approaches have produced substantial advances in biomedical signal analysis, they often treat cardiac acoustic events primarily as isolated classification targets rather than components of larger dynamical systems.

The present framework explores an alternative computational perspective centered on latent physiological topology and trajectory-based representation. Rather than modeling cardiac auscultation solely as static waveform classification, the framework investigates whether cardiac acoustic activity may be represented as movement through evolving physiological state-space. Within this interpretation, cardiac acoustic events become coordinates embedded within latent physiological manifolds, while sequential acoustic evolution becomes structured trajectory traversal through topology.

More specifically, the framework investigates whether physiological acoustic systems may exhibit recurrent manifold organization, directional transition persistence, cyclic trajectory behavior, distributed occupation structure, and partially stabilized latent state regions. The objective is not simply dimensionality reduction or visualization, but exploration of whether cardiac acoustic activity may contain measurable organizational structure when represented through nonlinear geometric state-space.

Conceptually, the framework extends ideas developed in the earlier *Bioacoustic Topology* project into the domain of biomedical acoustic systems. However, the present work remains exploratory and computational rather than clinically validated. The project does not attempt diagnostic inference, pathological classification, or physiologically validated latent topology. Instead, it proposes a systems-oriented computational perspective in which physiological acoustic activity may be investigated as dynamically evolving structure embedded within latent geometric manifolds.

2. Research Objectives

The present framework investigates whether cardiac acoustic activity may exhibit measurable latent organization when modeled as dynamical traversal through physiological state-space

rather than as isolated waveform events. More specifically, the project explores whether segmented cardiac acoustic events can be embedded into coherent latent physiological manifolds exhibiting recurrent organizational structure and identifiable trajectory dynamics.

A central objective of the framework is to investigate whether cardiac acoustic variability produces partially stabilized latent state regions and recurrent manifold occupation behavior within physiological topology. The project additionally examines whether sequential cardiac acoustic evolution generates structured transition pathways, directional manifold traversal, and dynamically distinguishable trajectory configurations.

The framework further investigates whether anomalous physiological acoustic events may be identifiable through latent trajectory dynamics, including manifold isolation, elevated traversal intensity, and abrupt directional transition behavior. Beyond Euclidean latent embedding, the project also explores whether cardiac acoustic activity may exhibit partially cyclic organizational structure when represented through recurrence-preserving toroidal geometry and orbital phase analysis.

More broadly, the framework evaluates whether topology-oriented computational visualization systems—including latent manifolds, physiological trajectories, orbital embeddings, anomaly topology, and holographic density rendering—may provide interpretable exploratory interfaces for biomedical acoustic signal analysis.

3. Dataset and Signal Acquisition

The present framework utilizes a subset of the CirCor DigiScope Phonocardiogram Dataset distributed through PhysioNet. The dataset contains pediatric cardiac auscultation recordings collected across multiple anatomical acquisition locations, including the aortic valve, mitral valve, pulmonary valve, and tricuspid valve regions. These recordings provide a suitable foundation for exploratory physiological acoustic analysis due to their recurrent cyclic structure, heterogeneous acoustic morphology, and clinically relevant signal variability.

For the purposes of the present exploratory study, a controlled subset of 40 waveform recordings was analyzed. The recordings were processed at a sampling rate of 4000 Hz and subsequently transformed into computational physiological acoustic representations suitable for latent manifold analysis.

Each waveform underwent a sequential computational processing pipeline consisting of acoustic segmentation, cardiac event extraction, multidimensional feature-space construction, latent manifold embedding, and physiological trajectory reconstruction. This processing framework enabled transformation of continuous cardiac acoustic recordings into structured latent physiological state-space representations suitable for dynamical topology analysis.

Importantly, the dataset subset was used for exploratory computational modeling rather than diagnostic evaluation or clinical validation. The objective of the framework is therefore not pathological inference, but investigation of whether cardiac acoustic systems may exhibit measurable latent organizational structure when represented through nonlinear physiological topology.

4. Computational Methodology

4.1 Physiological Acoustic Segmentation

Continuous cardiac acoustic waveforms were first transformed into discrete computational analysis units using onset-based acoustic segmentation. The segmentation framework identifies localized energetic transitions within the waveform and partitions the continuous physiological signal into temporally bounded cardiac acoustic events suitable for subsequent feature extraction and latent topology analysis.

This transformation is conceptually important because it enables reconstruction of physiological acoustic activity as sequential movement through latent state-space rather than as a single continuous waveform. Within the resulting framework, segmented cardiac acoustic events function as the fundamental computational units from which physiological trajectories, manifold organization, and transition dynamics are reconstructed.

The segmentation process additionally enables comparison of local acoustic morphology across temporally distinct physiological events while preserving sequential ordering throughout the recording. These segmented units therefore serve as the basis for multidimensional feature-space construction and dynamical manifold traversal analysis.

Importantly, the segmented events should be interpreted as computational acoustic analysis windows rather than clinically validated cardiac phases or physiologically discrete biological

states. The segmentation framework is designed primarily to support exploratory latent topology reconstruction rather than direct physiological annotation.

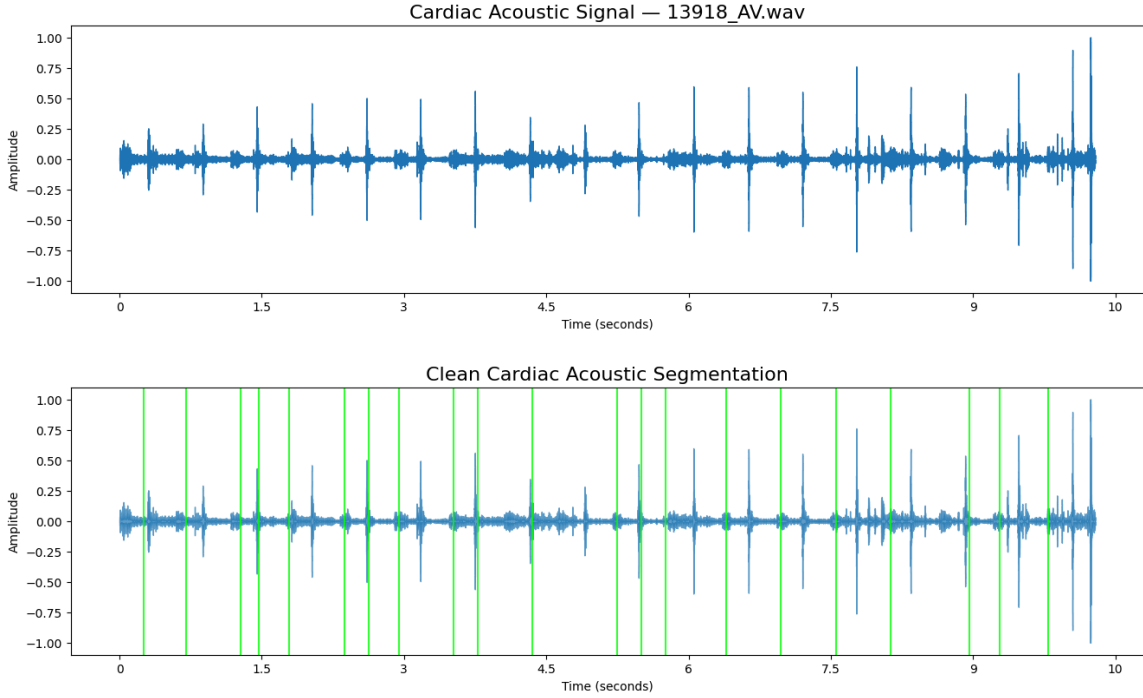


Figure 1. Cardiac acoustic waveform and computational segmentation structure. The continuous phonocardiogram signal is partitioned into discrete cardiac acoustic analysis windows using onset-based acoustic detection. Red segmentation markers indicate detected physiological acoustic event boundaries used for latent manifold reconstruction.

4.2 Physiological Feature Extraction

Following segmentation, each cardiac acoustic event was transformed into a multidimensional physiological descriptor vector designed to approximate local acoustic morphology within latent feature-space. The resulting representation integrates spectral, temporal, and energetic acoustic information in order to enable comparison of physiological acoustic structure across sequential cardiac events.

The feature space combines multiple complementary descriptor classes, including Mel-frequency cepstral coefficients (MFCCs), spectral centroid, spectral bandwidth, RMS energy, zero-crossing rate, and spectral rolloff. Collectively, these descriptors approximate variations in local frequency organization, energetic intensity, spectral distribution, and temporal acoustic behavior throughout the physiological signal.

MFCC representations were used to capture compact spectral envelope structure, while additional spectral descriptors provided estimates of acoustic concentration, bandwidth distribution, and energetic transition characteristics. RMS energy and zero-crossing behavior further contributed information related to local signal intensity and waveform complexity. Together, these descriptors generated a computational physiological feature-space suitable for manifold learning and trajectory reconstruction.

The resulting physiological feature matrix contained 21 segmented cardiac acoustic events represented across 18 acoustic feature dimensions. This transformation effectively converted the continuous cardiac waveform into a structured multidimensional physiological state-space capable of supporting latent manifold embedding, recurrent state discovery, and dynamical topology analysis.

4.3 Latent Physiological Manifold Construction

The standardized physiological feature matrix was projected into a three-dimensional latent manifold using Uniform Manifold Approximation and Projection (UMAP), a nonlinear dimensionality reduction framework designed to preserve local relational structure within high-dimensional data. The objective of this embedding process was not merely visual compression, but reconstruction of a latent physiological topology capable of approximating underlying acoustic organization across sequential cardiac events.

Within the resulting latent manifold, acoustically similar cardiac events occupy neighboring regions of physiological state-space, while acoustically distinct events become increasingly separated throughout the topology. Recurrent physiological structures additionally form partially concentrated manifold occupation regions, suggesting non-uniform latent organization within the cardiac acoustic signal.

Importantly, temporal indexing of sequential cardiac events transforms the manifold from a static geometric embedding into a dynamical trajectory system. Within this framework, sequential cardiac acoustic evolution generates continuous latent trajectory pathways traversing physiological topology through time. This enables investigation not only of local acoustic similarity, but also of recurrent manifold traversal behavior, directional transition persistence, and distributed physiological state-space organization.

The latent manifold should therefore be interpreted as a computational approximation of physiological acoustic organization rather than a purely visual dimensionality reduction

system. Its primary function within the framework is to provide a geometrically structured state-space suitable for trajectory reconstruction, recurrent state analysis, anomaly detection, and cyclic topology exploration.

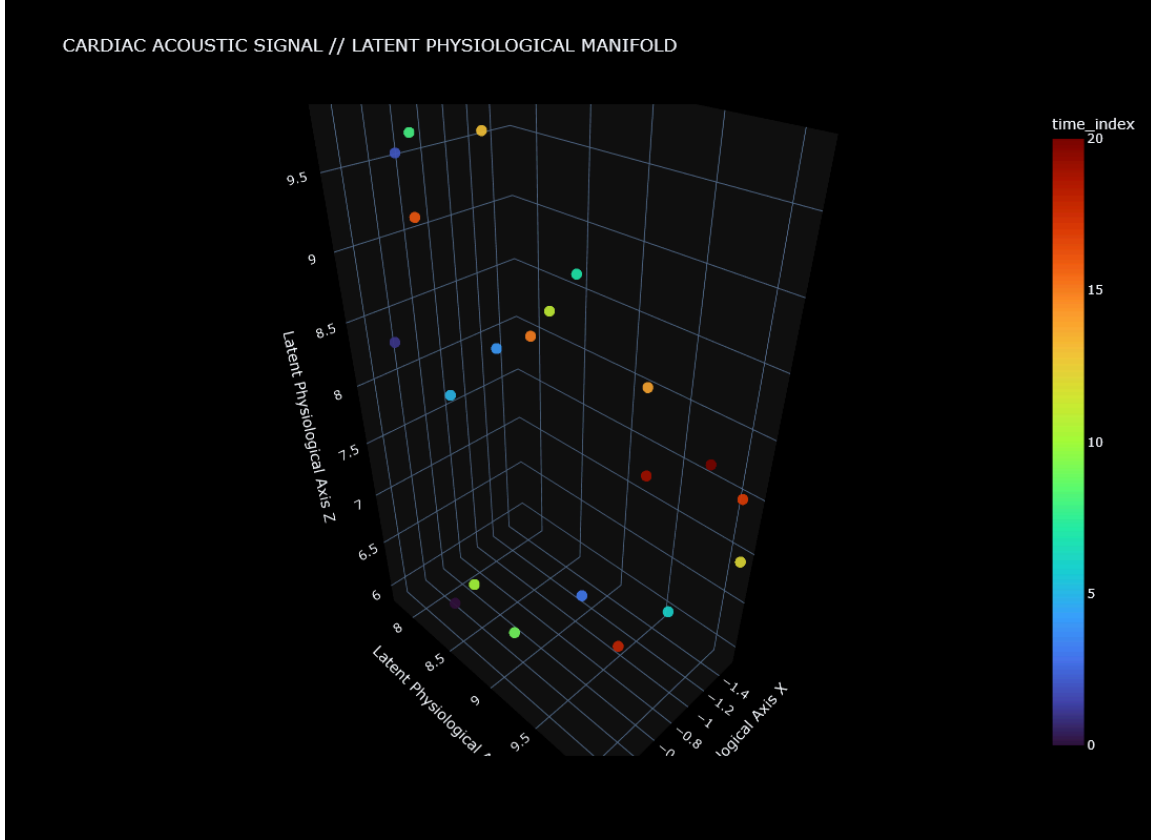


Figure 2. Three-dimensional latent physiological manifold generated using UMAP projection of multidimensional cardiac acoustic feature-space. Sequential cardiac acoustic events form structured trajectory pathways throughout latent physiological topology. Neighboring manifold regions correspond to acoustically similar physiological events.

4.4 Physiological State Discovery

Unsupervised K-Means clustering was applied within latent physiological topology in order to identify recurrent acoustic state regions distributed throughout the manifold. By partitioning latent physiological space into locally concentrated occupation zones, the framework investigates whether sequential cardiac acoustic activity exhibits partially recurrent organizational structure rather than uniformly random manifold traversal.

The clustering framework identified four recurrent physiological acoustic states distributed across latent topology. These states correspond to recurrent manifold occupation regions generated through similarities in multidimensional acoustic feature structure and trajectory organization. Several states exhibited partial spatial concentration and repeated trajectory visitation behavior, suggesting recurrent latent physiological organization within the analyzed recordings.

Within the broader framework, these state regions are interpreted as computational approximations of recurrent physiological acoustic structure, latent manifold occupation zones, or partially stabilized acoustic configurations embedded within state-space. Importantly, the identified states should not be interpreted as clinically validated physiological categories, pathological labels, or biologically discrete cardiac phases. Rather, they function as exploratory computational constructs generated through unsupervised latent topology analysis.

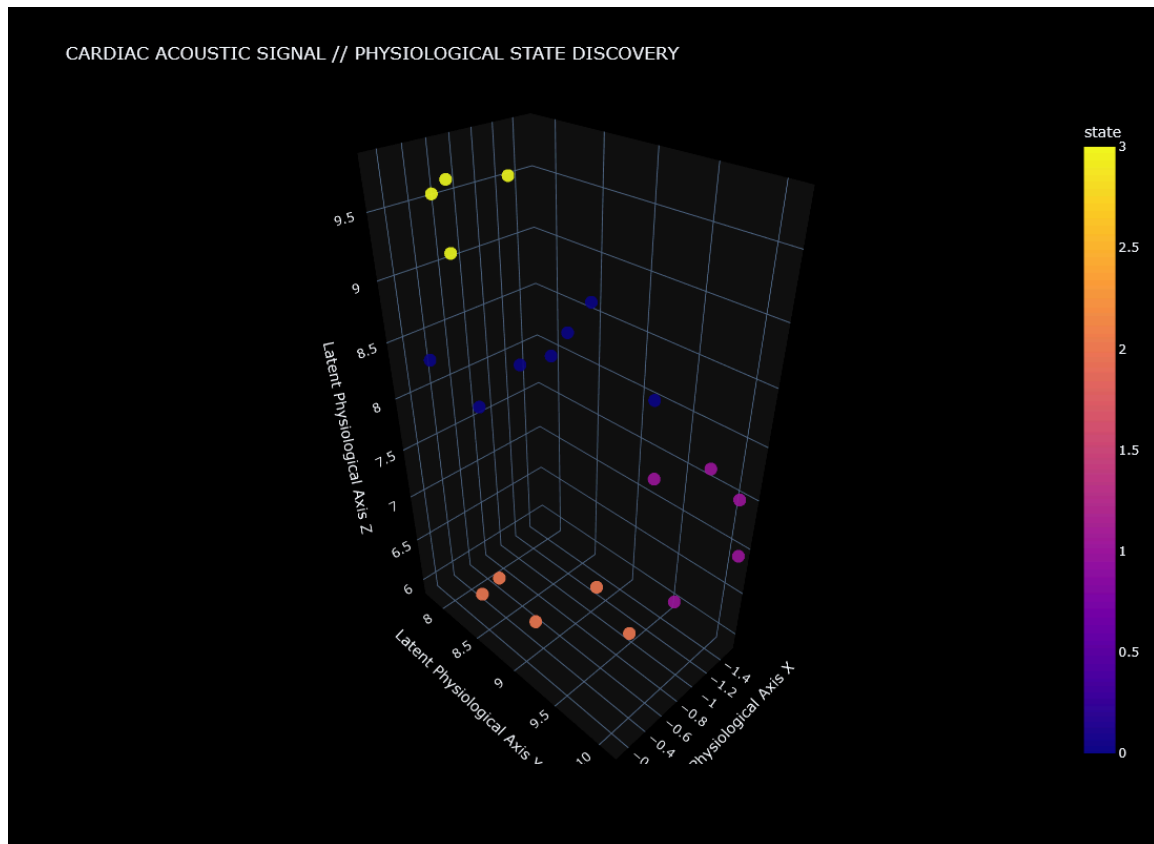


Figure 3. Recurrent physiological acoustic states identified through unsupervised clustering within latent physiological topology. Distinct colors represent computationally inferred latent physiological state regions distributed throughout manifold space.

4.5 Transition Dynamics and Probabilistic State Modeling

Sequential cardiac acoustic events were reconstructed as continuous latent trajectories traversing physiological state-space through time. Within this framework, individual cardiac acoustic events function as temporally ordered nodes embedded within latent topology, while successive event transitions generate directional manifold pathways approximating physiological acoustic evolution.

To investigate recurrent organizational structure within these trajectories, transition probabilities were estimated between discovered physiological acoustic states using a first-order probabilistic transition framework. This approach enables analysis of recurrent transition pathways, state persistence behavior, directional manifold traversal, and broader trajectory organization throughout latent physiological topology.

The resulting transition structure revealed several recurrent state pathways exhibiting elevated transition persistence relative to other manifold trajectories. Certain state regions demonstrated repeated directional traversal behavior and recurrent transition circulation, suggesting partially stabilized dynamical organization rather than uniformly random movement through latent physiological space.

Importantly, the probabilistic transition framework should be interpreted as a computational approximation of sequential physiological organization rather than a mechanistic physiological model. The objective is not prediction of biological cardiac behavior, but investigation of whether latent physiological topology exhibits measurable recurrent transition structure when reconstructed as a dynamical trajectory system.

4.6 Physiological Anomaly Detection

Trajectory-based anomaly scoring was applied in order to identify dynamically unusual cardiac acoustic events occupying atypical regions within latent physiological topology. Rather than relying solely on waveform amplitude irregularities, the anomaly framework evaluates physiological acoustic events according to their relational position and dynamical behavior within reconstructed latent state-space.

The anomaly analysis integrates multiple dynamical quantities, including latent-space isolation, trajectory displacement magnitude, velocity dynamics, and acceleration behavior. Collectively, these measures approximate the degree to which individual cardiac acoustic events

diverge from recurrent manifold traversal structure and established latent physiological organization.

Within the analyzed recordings, the framework identified three anomalous physiological acoustic events exhibiting elevated dynamical deviation relative to surrounding trajectory structure. Several of these anomalous events additionally coincided with regions of increased latent trajectory displacement and abrupt directional transition behavior, suggesting locally unusual manifold traversal dynamics.

Importantly, anomaly interpretation remains exploratory and computational rather than clinically diagnostic. The identified anomalies should not be interpreted as pathological findings or medically significant physiological abnormalities. Instead, they represent computationally unusual latent trajectory structures detected within the reconstructed physiological state-space framework.

4.7 Trajectory Velocity and Acceleration Estimation

Additional dynamical quantities were estimated directly from reconstructed latent physiological trajectories in order to characterize the intensity and variability of manifold traversal behavior throughout sequential cardiac acoustic evolution. These quantities extend the framework beyond static manifold organization by introducing explicit measures of latent dynamical movement within physiological state-space.

The estimated dynamical descriptors included latent trajectory velocity, acceleration, and local transition intensity. Latent velocity estimates approximate the magnitude of movement between successive physiological acoustic events within manifold topology, thereby providing a measure of local trajectory displacement through state-space. Acceleration estimates additionally quantify abrupt changes in trajectory progression by measuring variation in latent traversal magnitude across consecutive transitions.

Together, these quantities enable investigation of whether cardiac acoustic evolution exhibits dynamically heterogeneous behavior, including abrupt manifold transitions, locally elevated traversal intensity, or regions of relative trajectory stabilization. Several trajectory intervals exhibited elevated velocity and acceleration behavior, suggesting non-uniform latent physiological dynamics throughout the analyzed recordings.

Importantly, these quantities should be interpreted as computational descriptors of latent trajectory behavior rather than direct physiological measurements. Their primary function

within the framework is to support exploratory dynamical topology analysis and anomaly characterization within reconstructed physiological state-space.

4.8 Experimental Toroidal Physiological Topology

To investigate whether cyclic manifold geometry may better preserve recurrent physiological organization, latent trajectories were additionally projected onto an experimental toroidal manifold representation. The motivation for this transformation arises from the inherently recurrent and cyclic character of physiological acoustic systems, which may not always be optimally represented within purely Euclidean latent topology.

Within the toroidal representation, angular coordinates approximate cyclic physiological progression across latent state-space. This geometric transformation introduces periodic continuity into the manifold, allowing recurrent trajectory returns to become geometrically compressed and partially cyclic traversal behavior to emerge more coherently within the reconstructed topology. As a result, orbital recurrence pathways and repeated manifold circulation structures become more visually interpretable than within conventional linear embedding space alone.

Conceptually, the toroidal manifold functions as a recurrence-preserving computational projection designed to emphasize cyclic organization within physiological trajectory systems. The objective is not to propose literal toroidal physiological geometry, but rather to explore whether cyclic embedding frameworks may reveal latent recurrent structure that remains partially fragmented under standard Euclidean manifold representations.

Importantly, the toroidal embedding should therefore be interpreted as an exploratory computational abstraction rather than a physiologically validated geometric model. Its value lies primarily in supporting investigation of cyclic recurrence behavior, orbital trajectory organization, and distributed manifold circulation within latent physiological topology.

4.9 Orbital Phase Synchronization Analysis

Orbital phase statistics were estimated in order to investigate cyclic recurrence behavior within toroidal physiological topology. Following toroidal projection, latent physiological trajectories may be interpreted as evolving through partially cyclic orbital regions, enabling exploratory analysis of recurrent phase organization and manifold circulation behavior.

Within this framework, orbital phase estimates approximate the relative angular position of physiological acoustic events throughout toroidal state-space. Sequential variation in orbital phase therefore provides a computational representation of cyclic trajectory evolution, recurrent orbital occupation, and directional manifold circulation through time.

The analysis specifically investigates whether physiological acoustic states exhibit recurrent orbital visitation behavior, partially cyclic traversal pathways, and localized phase persistence across successive trajectory intervals. Several trajectory regions demonstrated non-uniform orbital evolution and partially recurrent circulation structure, suggesting that certain physiological acoustic configurations repeatedly occupy neighboring cyclic manifold regions.

Importantly, orbital synchronization analysis should be interpreted as an exploratory computational approximation of cyclic latent organization rather than evidence of physiological oscillatory synchronization mechanisms. The framework does not claim biologically validated orbital dynamics, but instead explores whether recurrence-preserving topology may reveal measurable cyclic organizational structure within latent physiological state-space.

4.10 Holographic Physiological Density Fields

Latent physiological topology was additionally rendered as volumetric holographic density fields in order to visualize distributed manifold occupation behavior throughout reconstructed physiological state-space. Rather than emphasizing isolated trajectory nodes alone, the holographic representation treats latent physiological organization as a continuous distributed structure embedded within geometric topology.

Within this framework, density fields approximate recurrent manifold occupation, physiological state concentration, distributed latent organization, and energetic trajectory circulation behavior across sequential cardiac acoustic evolution. Regions exhibiting repeated trajectory traversal or concentrated state occupation generate higher-density manifold structures, while dynamically isolated regions remain comparatively sparse within latent topology.

The holographic representation additionally enables visualization of overlap relationships between recurrent physiological states and transitional manifold pathways. As a result, the framework emphasizes systems-level organization and distributed acoustic topology rather than purely discrete event classification.

Conceptually, the density rendering draws inspiration from volumetric dynamical systems visualization and computational field modeling. Importantly, however, the resulting density structures should not be interpreted as physical physiological measurements, energetic tissue properties, or biological field dynamics. Instead, they represent computational approximations of recurrent manifold occupation behavior within latent physiological topology.

5. Results

5.1 Latent Physiological Manifold Organization

Projection of the physiological acoustic feature space into latent topology revealed coherent organizational structure among sequential cardiac acoustic events. Within the resulting manifold, acoustically similar events occupied neighboring latent regions, while acoustically distinct events became increasingly separated throughout physiological state-space. This spatial organization suggests that the multidimensional feature representation preserves meaningful local relational structure across the analyzed cardiac acoustic recordings.

Several latent regions additionally exhibited partial manifold concentration behavior, indicating recurrent occupation of specific physiological acoustic configurations. By contrast, a smaller number of events occupied comparatively isolated manifold positions, suggesting the presence of acoustically atypical or dynamically distinct trajectory structures within the broader topology.

Importantly, temporal reconstruction transformed the manifold from a static geometric embedding into a dynamical physiological trajectory system. Sequential cardiac acoustic events generated continuous latent traversal pathways exhibiting directional organization rather than uniformly random manifold movement. Several trajectory intervals repeatedly traversed neighboring latent corridors, suggesting partially recurrent manifold circulation behavior throughout physiological state-space.

Collectively, these observations support interpretation of the latent manifold as a computational approximation of physiological acoustic organization rather than a purely visual dimensionality reduction projection. The reconstructed topology therefore provides a structured geometric framework for investigating recurrent physiological states, anomaly dynamics, and cyclic trajectory organization.

5.2 Recurrent Physiological Acoustic States

Unsupervised clustering within latent physiological topology identified four recurrent physiological acoustic states distributed throughout the reconstructed manifold. These states emerged as partially differentiated regions of latent occupation generated through similarities in multidimensional acoustic feature structure and sequential trajectory organization.

The identified states exhibited varying levels of spatial concentration, trajectory persistence, and recurrent manifold occupation behavior. Several states formed comparatively compact latent regions characterized by repeated trajectory visitation, while others displayed broader spatial dispersion and more variable transition behavior throughout physiological state-space.

This variation in manifold organization suggests that cardiac acoustic evolution may contain both relatively stable recurrent physiological configurations and more dynamically transitional acoustic structures. Certain trajectory pathways repeatedly returned to neighboring latent regions, indicating partially recurrent manifold circulation behavior across sequential cardiac acoustic progression.

Importantly, these recurrent physiological states should be interpreted as computationally inferred latent structures rather than clinically validated physiological categories or biological cardiac phases. Their significance within the framework lies primarily in supporting exploratory investigation of recurrent organization, latent occupation dynamics, and physiological trajectory topology.

5.3 Transition Dynamics

The physiological transition matrix revealed directional organization between recurrent latent physiological states throughout sequential cardiac acoustic evolution. Rather than exhibiting uniformly distributed transition behavior, several state pathways demonstrated elevated recurrence probabilities and partially stabilized traversal dynamics within latent physiological topology.

Most notably, transitions from State 2 to State 0 exhibited strong transition persistence, suggesting recurrent directional movement between neighboring manifold occupation regions. Similarly, transitions from State 3 to State 2 demonstrated repeated directional traversal behavior, while recurrent pathways between State 0 and State 3 formed partially circulating manifold structures suggestive of recurrent trajectory organization.

These recurrent transition pathways indicate that latent physiological evolution throughout the analyzed recordings is not entirely random, but instead exhibits identifiable directional structure within reconstructed state-space. Several trajectory intervals repeatedly traversed similar latent corridors, producing partially conserved transition architecture across sequential cardiac acoustic events.

Importantly, the transition matrix should be interpreted as a computational approximation of latent sequential organization rather than a mechanistic physiological model. Nevertheless, the observed recurrence structure supports the broader interpretation that physiological acoustic systems may exhibit measurable trajectory organization when reconstructed through latent manifold dynamics.

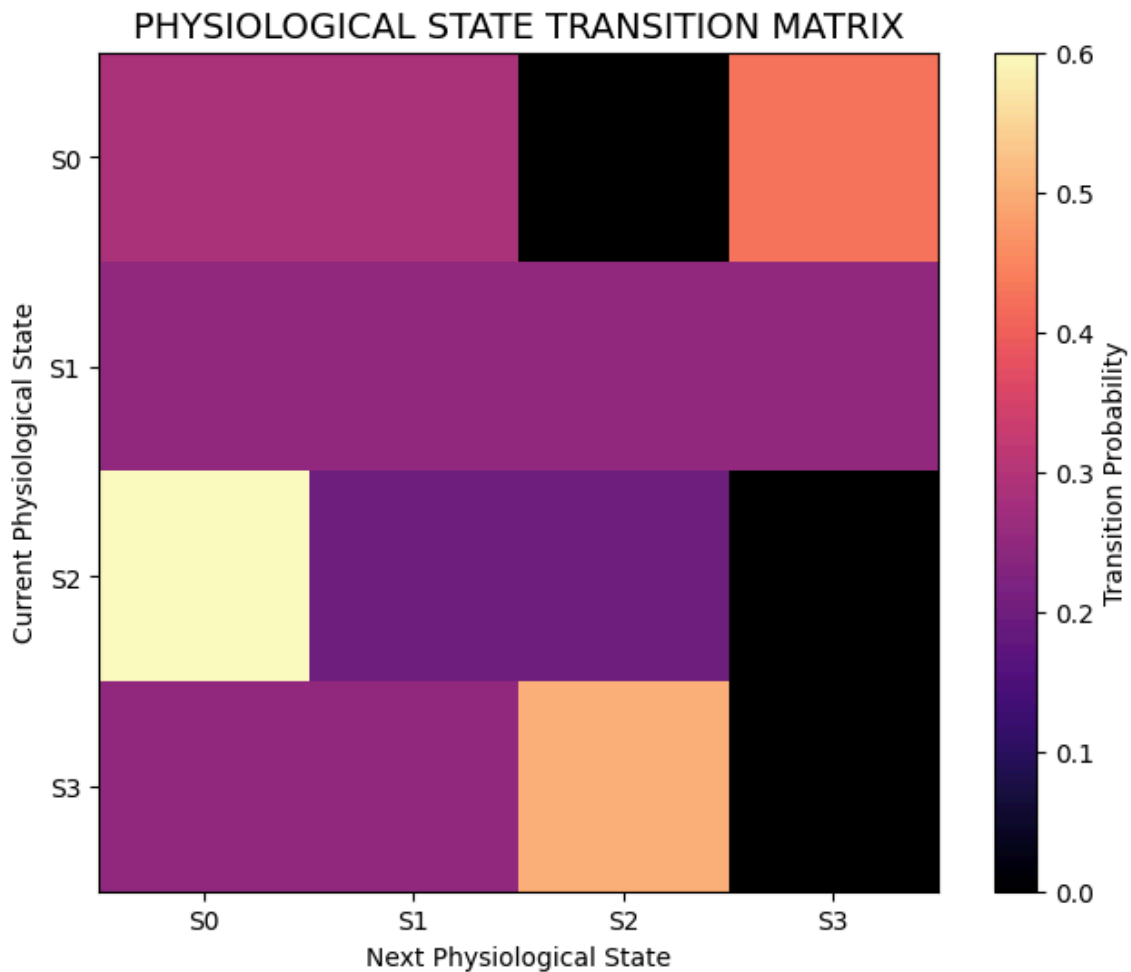


Figure 4. Probabilistic physiological state transition matrix illustrating directional traversal behavior between recurrent latent physiological states. Elevated transition probabilities indicate partially stabilized trajectory pathways within reconstructed physiological topology.

5.4 Physiological Anomaly Structure

Trajectory-based anomaly analysis identified three dynamically unusual physiological acoustic events occupying atypical regions within latent physiological topology. These anomalous events diverged from recurrent manifold traversal structure and appeared comparatively isolated relative to surrounding physiological trajectory organization.

Several anomalies occupied spatially distinct latent manifold regions associated with elevated trajectory displacement behavior and reduced local manifold density. In addition to their relative geometric isolation, the identified anomalous events frequently coincided with regions exhibiting increased latent velocity and abrupt acceleration dynamics, suggesting locally intensified trajectory reconfiguration within physiological state-space.

The convergence between anomaly localization, elevated traversal magnitude, and abrupt directional transition behavior suggests that dynamically unusual physiological acoustic events may correspond to transient deviations from recurrent manifold organization. Rather than representing stable recurrent occupation regions, these events appear associated with localized perturbations or transitional latent trajectory behavior.

Importantly, anomaly interpretation remains exploratory and computational rather than clinically diagnostic. The framework does not infer pathological significance or medically validated physiological abnormality from the identified events. Instead, the anomaly structures function as computational indicators of dynamically unusual latent physiological organization within the reconstructed state-space framework.

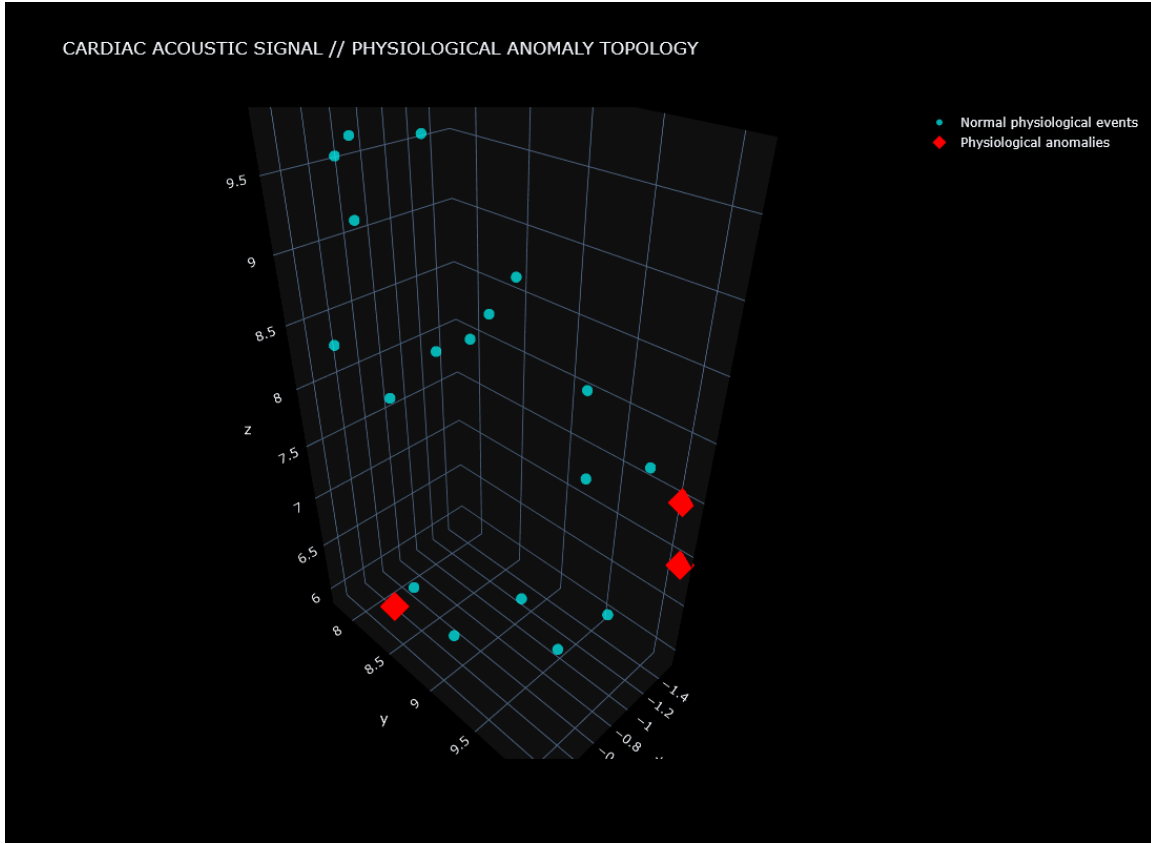


Figure 5. Trajectory-based physiological anomaly topology. Dynamically unusual cardiac acoustic events occupy isolated latent manifold regions and exhibit elevated displacement behavior relative to recurrent physiological trajectory organization.

5.5 Dynamical Trajectory Behavior

Estimation of latent trajectory velocity and acceleration revealed substantial variation in manifold traversal intensity throughout sequential cardiac acoustic evolution. Rather than progressing through physiological state-space at uniform dynamical rates, the reconstructed trajectories exhibited periods of relative stability interspersed with intervals of abrupt latent displacement and elevated traversal activity.

Several trajectory intervals demonstrated increased latent velocity, indicating rapid movement between neighboring manifold regions and intensified state-space traversal behavior. Additional intervals exhibited abrupt acceleration spikes corresponding to sudden changes in directional progression and localized trajectory reconfiguration within latent topology. These acceleration events frequently coincided with transitional manifold pathways and dynamically unusual acoustic structures identified through anomaly analysis.

The resulting dynamical profile suggests that physiological acoustic evolution throughout the analyzed recordings is characterized by heterogeneous trajectory organization rather than smooth continuous manifold traversal. Certain regions of latent topology appear comparatively stable and recurrent, while others function as higher-intensity transition corridors associated with abrupt dynamical movement through state-space.

Importantly, velocity and acceleration estimates should be interpreted as computational descriptors of latent trajectory behavior rather than direct physiological measurements. Their significance within the framework lies primarily in supporting exploratory investigation of dynamical organization, transition intensity, and manifold traversal structure within reconstructed physiological topology.

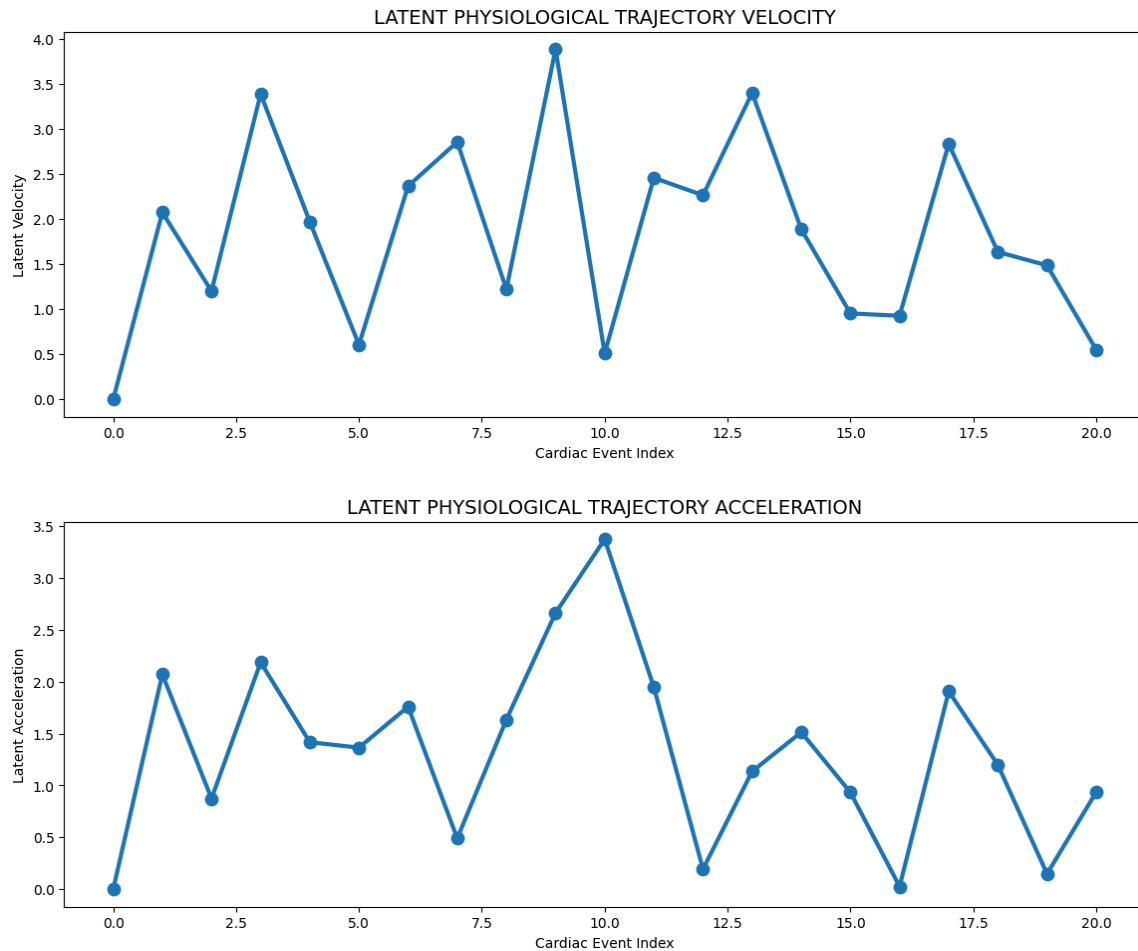


Figure 6. Latent physiological trajectory velocity and acceleration dynamics across sequential cardiac acoustic evolution. Elevated traversal magnitude and abrupt acceleration transitions indicate dynamically heterogeneous manifold traversal behavior.

5.6 Toroidal Physiological Topology

Projection of latent physiological trajectories onto the experimental toroidal manifold introduced cyclic continuity into reconstructed physiological topology. Within this recurrence-preserving geometric representation, sequential cardiac acoustic evolution exhibited more coherent cyclic organization than was immediately apparent within the original Euclidean manifold embedding.

Several recurrent trajectory returns became geometrically compressed within toroidal space, allowing partially cyclic traversal pathways and orbital recurrence behavior to emerge more clearly throughout latent physiological topology. Repeated manifold visitation patterns additionally generated partially stabilized circulation structures suggestive of recurrent cyclic organization across sequential physiological acoustic evolution.

The toroidal representation also reduced a degree of apparent fragmentation previously observed within Euclidean latent topology. Trajectory regions that appeared comparatively dispersed under linear embedding became more continuously organized under cyclic manifold projection, suggesting that recurrence-preserving geometry may better approximate certain forms of latent physiological organization within cyclic acoustic systems.

Importantly, the toroidal manifold should be interpreted as an exploratory computational abstraction rather than a physiologically validated geometric model. Nevertheless, the resulting topology demonstrates that cyclic manifold representations may provide useful frameworks for investigating recurrent physiological organization, orbital trajectory behavior, and distributed manifold circulation within latent state-space.

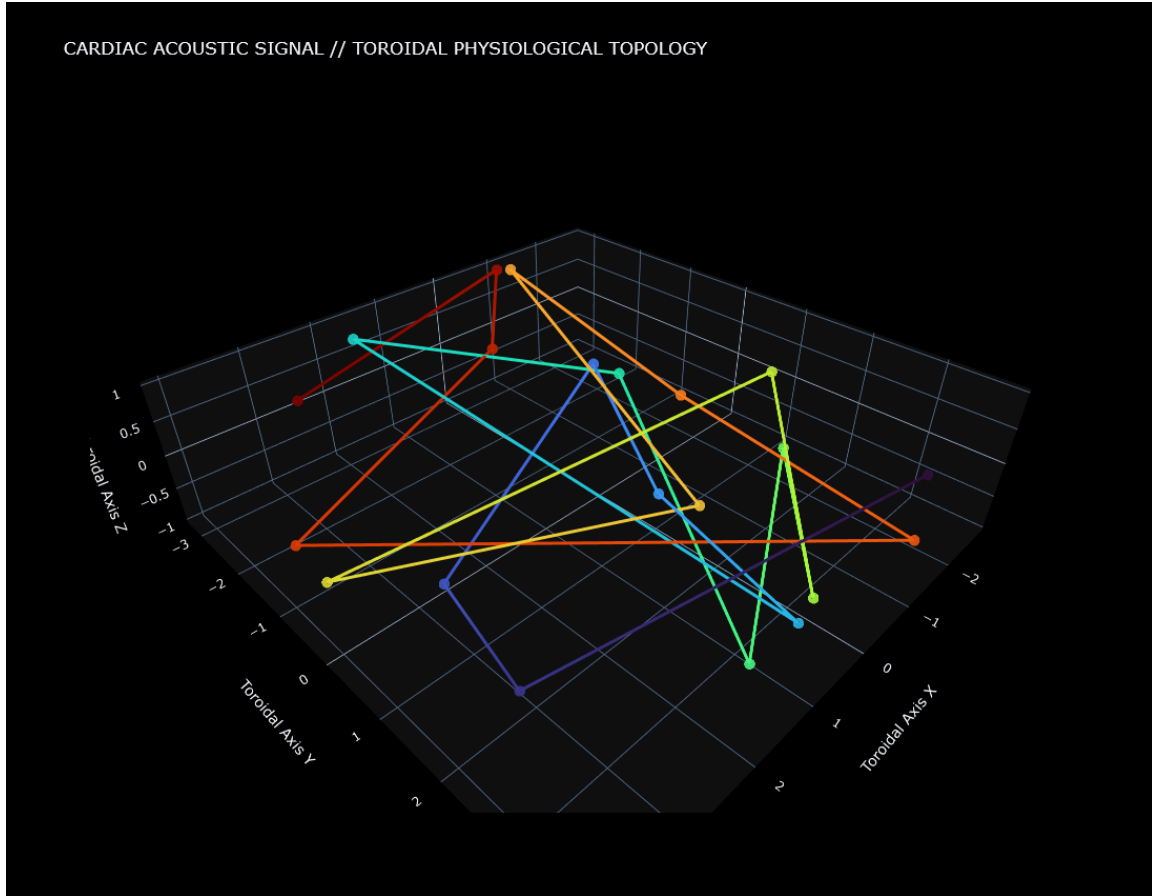


Figure 7. Experimental toroidal physiological topology generated through cyclic manifold projection of latent cardiac acoustic trajectories. Recurrent trajectory returns and orbital circulation behavior become more visually coherent within recurrence-preserving toroidal geometry.

5.7 Orbital Phase Recurrence

Orbital phase analysis revealed non-uniform cyclic traversal behavior throughout toroidal physiological topology. Rather than evolving through latent orbital space at constant phase progression rates, the reconstructed trajectories exhibited recurrent orbital occupation patterns, localized phase persistence, and intervals of abrupt cyclic reconfiguration.

Several physiological trajectory regions repeatedly occupied neighboring orbital sectors within toroidal topology, suggesting partially stabilized cyclic organization and recurrent manifold circulation behavior. Certain orbital pathways additionally demonstrated relative phase persistence across successive trajectory intervals, indicating locally recurrent traversal structure within cyclic physiological state-space.

By contrast, several trajectory intervals exhibited abrupt orbital phase transitions associated with elevated latent acceleration dynamics and intensified manifold displacement behavior. These rapid phase changes frequently coincided with dynamically unusual trajectory regions identified through anomaly analysis, suggesting localized disruption or reorganization of recurrent cyclic traversal patterns.

Importantly, orbital phase recurrence should be interpreted as an exploratory computational descriptor of cyclic manifold organization rather than evidence of physiologically validated oscillatory synchronization mechanisms. The framework instead investigates whether recurrence-preserving topology may reveal measurable cyclic structure within latent physiological trajectories reconstructed from sequential cardiac acoustic activity.

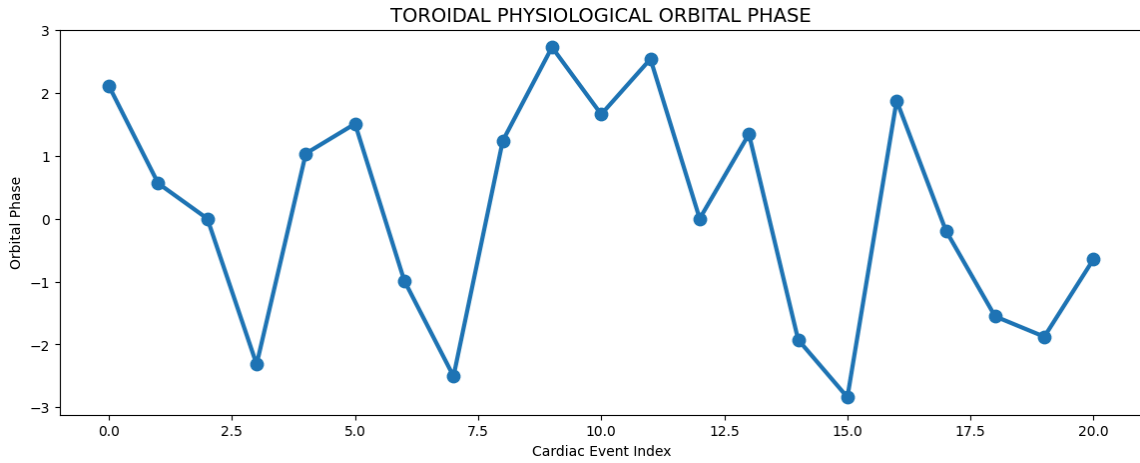


Figure 8. Orbital phase evolution throughout toroidal physiological topology. Non-uniform cyclic traversal behavior and recurrent orbital occupation patterns reveal partially stabilized cyclic manifold organization.

5.8 Holographic Physiological Density Fields

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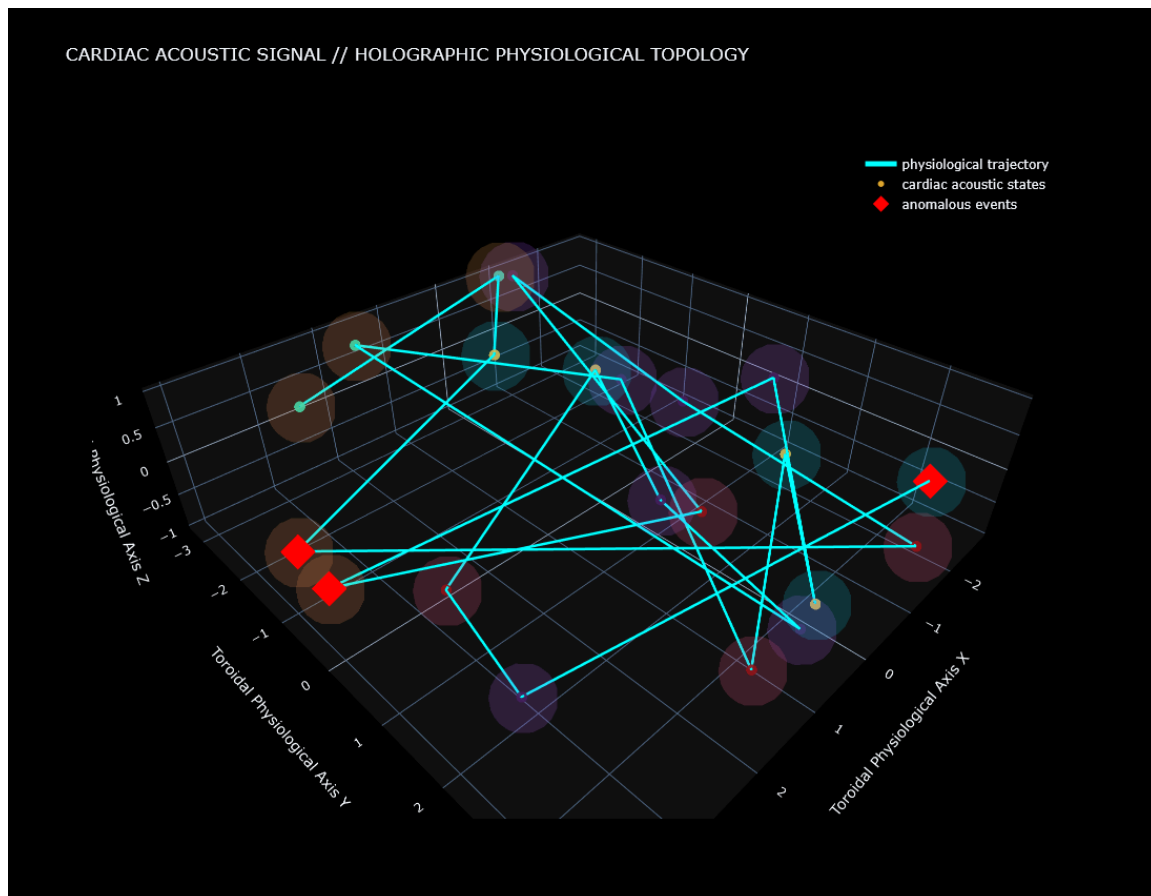


Figure 9. Holographic physiological density topology. Volumetric latent density rendering highlights distributed manifold occupation regions, recurrent trajectory circulation behavior, and concentrated physiological state-space organization throughout reconstructed cardiac acoustic topology.

6. Discussion

The present framework investigated whether cardiac acoustic activity may exhibit measurable latent organization when modeled as movement through physiological state-space rather than as isolated waveform events. Across the analyzed recordings, several recurrent structural properties emerged, including coherent latent manifold organization, recurrent physiological acoustic states, directional transition persistence, anomaly structures, cyclic manifold recurrence behavior, orbital phase organization, and distributed holographic occupation regions. Collectively, these observations suggest that physiological acoustic systems may exhibit substantially richer latent organization than is immediately visible through conventional waveform inspection alone.

A central conceptual shift within the framework is the transition from static acoustic analysis toward dynamical trajectory modeling. Traditional computational approaches frequently emphasize waveform categorization or diagnostic classification, often treating cardiac acoustic events as independent analysis targets. By contrast, the present system reconstructs cardiac acoustic evolution as structured traversal through latent physiological topology. Within this interpretation, individual cardiac acoustic events become coordinates embedded within manifold geometry, transitions become dynamical pathways through state-space, and recurrent physiological behavior becomes organized manifold circulation across sequential trajectory evolution.

This trajectory-oriented perspective enables investigation of physiological acoustic systems through concepts more commonly associated with nonlinear dynamical systems and computational topology, including recurrent manifold occupation, directional traversal persistence, cyclic recurrence behavior, and distributed latent structure. The resulting framework therefore shifts emphasis away from isolated acoustic classification toward exploration of relational organization embedded within evolving physiological state-space.

The experimental toroidal embedding and orbital phase analysis further suggest that recurrence-preserving manifold geometry may reveal cyclic organizational structure not immediately apparent within conventional Euclidean latent representations. Similarly, holographic density rendering emphasizes distributed manifold occupation behavior rather

than discrete acoustic event isolation alone, supporting a more systems-oriented interpretation of physiological acoustic organization.

Importantly, the framework remains exploratory and computational rather than clinically validated. The project does not claim diagnostic capability, pathological classification, physiological mechanism discovery, or medically validated latent topology. Similarly, concepts such as orbital recurrence, cyclic manifold circulation, and partially stabilized physiological states should be interpreted as computational abstractions describing latent organizational behavior rather than direct biological mechanisms.

Instead, the broader contribution of the framework lies in proposing an alternative computational perspective in which physiological acoustic systems may be investigated through manifold topology, trajectory dynamics, cyclic recurrence, and distributed latent organization. Within this perspective, cardiac acoustic activity is explored not solely as a biomedical signal classification problem, but as a dynamically evolving geometric structure embedded within latent physiological state-space.

7. Limitations

The present framework should be interpreted as exploratory and hypothesis-generating rather than clinically validated. Although the reconstructed latent physiological topology revealed recurrent organizational structure throughout the analyzed recordings, several important methodological and interpretive limitations must be acknowledged.

7.1 Dataset Scope

The current analysis was conducted using a limited subset of cardiac acoustic recordings derived from the CirCor DigiScope Phonocardiogram Dataset. Consequently, the observed manifold structures, recurrent trajectory patterns, anomaly regions, and cyclic organizational behaviors remain strongly dependent on the specific recordings analyzed within the present exploratory framework.

As a result, the framework cannot determine whether the observed latent physiological structures generalize across broader populations, pathological categories, demographic variation, acquisition environments, or diverse clinical conditions. Similarly, the present analysis does not evaluate whether recurrent manifold organization remains stable across larger physiological recording distributions or clinically heterogeneous datasets.

Future work would therefore require substantially larger comparative physiological datasets, cross-recording validation, and integration of clinically annotated metadata in order to evaluate the robustness, reproducibility, and broader applicability of the reconstructed latent topology.

7.2 Segmentation Uncertainty

Cardiac acoustic segmentation within the present framework was performed computationally using onset-based acoustic detection rather than manually validated physiological annotation. Although this approach enables automated transformation of continuous waveforms into discrete computational analysis units, the resulting segmentation boundaries remain approximate and may not perfectly correspond to physiologically meaningful cardiac phases.

Variability in waveform morphology, local signal energy, recording quality, and acoustic transition behavior may additionally influence segmentation precision throughout the analyzed recordings. As a result, certain segmented events may represent partially merged physiological structures, while others may fragment continuous acoustic behavior into multiple computational windows.

Because the subsequent latent manifold construction, trajectory reconstruction, and dynamical topology analysis depend directly on the segmented event structure, segmentation uncertainty may propagate throughout later stages of the framework. The resulting latent topology should therefore be interpreted as a computational approximation of physiological acoustic organization rather than a definitive representation of discrete biological cardiac states.

Future work incorporating clinically annotated cardiac phase segmentation, multimodal physiological synchronization, or more advanced adaptive segmentation strategies may help improve alignment between computational acoustic events and underlying physiological structure.

7.3 Embedding Sensitivity

The reconstructed latent physiological manifold depends strongly on multiple computational design choices involved in feature-space construction and dimensionality reduction. Variations in acoustic descriptor selection, feature normalization strategies, manifold learning parameters, neighborhood configuration, and embedding dimensionality may substantially influence the resulting topology and trajectory organization.

In particular, the UMAP embedding process emphasizes preservation of local relational structure while approximating higher-dimensional organization within lower-dimensional latent space. Consequently, alternative parameter configurations may produce different manifold geometries, transition pathways, occupation densities, and apparent state relationships throughout physiological topology.

The resulting manifold should therefore not be interpreted as a unique or objectively correct representation of physiological acoustic organization. Instead, it represents one computational approximation of latent relational structure generated under a specific embedding configuration and feature-space formulation.

Similarly, the observed recurrent state regions, cyclic traversal patterns, and anomaly structures may vary under alternative manifold learning approaches, including different nonlinear embedding algorithms or higher-dimensional latent representations. Future work involving comparative embedding analysis, stability estimation, and topological consistency evaluation may therefore help clarify which latent organizational properties remain robust across varying computational configurations.

7.4 Clinical Constraints

The present framework does not incorporate diagnostic inference, pathological validation, or clinically validated physiological labeling. Although the reconstructed latent topology revealed recurrent organizational structure throughout the analyzed cardiac acoustic recordings, the framework was designed primarily for exploratory computational modeling rather than medical decision-making or clinical interpretation.

Consequently, the identified physiological states, anomaly regions, transition pathways, and cyclic manifold structures should not be interpreted as clinically validated indicators of cardiac pathology or physiological dysfunction. Similarly, the framework does not establish causal physiological mechanisms underlying the observed latent organizational behavior.

Several conceptual elements introduced throughout the analysis—including toroidal geometry, orbital recurrence behavior, attractor-like manifold occupation, and cyclic trajectory persistence—should therefore be understood as computational abstractions describing relational organization within latent state-space rather than direct physiological mechanisms operating within biological cardiac systems.

The broader objective of the framework is not diagnostic classification, but investigation of whether physiological acoustic systems may exhibit measurable latent organization when reconstructed through nonlinear manifold learning and dynamical trajectory analysis. Clinical interpretation, physiological validation, and pathological relevance would require substantially larger annotated datasets, expert biomedical evaluation, and rigorous comparative validation studies beyond the scope of the present exploratory work.

8. Future Directions

Future research may substantially extend the present framework through application to larger-scale physiological datasets incorporating broader demographic diversity, clinically annotated recordings, pathological variation, and multimodal biomedical acquisition environments. Comparative analysis across heterogeneous physiological populations may help clarify whether recurrent latent organizational structures remain stable across differing cardiac conditions, recording contexts, and physiological states.

Additional extensions may involve integration of multimodal physiological information, including electrocardiographic signals, respiratory dynamics, hemodynamic measurements, or temporally synchronized biomedical sensor streams. Such approaches may enable investigation of how latent physiological topology evolves across interacting biological systems rather than within isolated acoustic recordings alone.

From a computational perspective, several advanced modeling directions remain particularly promising. Persistent homology and topological data analysis may provide more rigorous characterization of recurrent manifold structures, cyclic occupation regions, and higher-order latent organization within physiological state-space. Similarly, graph neural network architectures and transformer-based sequence embeddings may support more sophisticated modeling of long-range physiological trajectory dependencies and recurrent transition dynamics.

Future work may also investigate streaming physiological topology systems capable of real-time manifold reconstruction, continuous trajectory visualization, and interactive biomedical state-space exploration. Such systems could potentially support exploratory computational interfaces for biomedical signal analysis and physiological monitoring applications.

The exploratory toroidal embedding introduced throughout the present framework additionally represents only one possible recurrence-preserving geometric representation. Further investigation of hyperspherical embeddings, hyperbolic physiological manifolds, recurrent loop geometries, and higher-dimensional cyclic topologies may reveal alternative approaches for preserving latent trajectory organization within recurrent physiological systems.

More broadly, the framework contributes toward an emerging interdisciplinary research direction situated at the intersection of biomedical signal analysis, manifold learning, nonlinear dynamical systems, geometric machine learning, computational topology, and physiological trajectory modeling. Within such a perspective, physiological acoustic systems may be investigated not merely as biomedical signals, but as dynamically evolving structures embedded within latent geometric state-space.

9. Conclusion

This report presented an exploratory computational framework for modeling cardiac acoustic signals as movement through latent physiological state-space rather than as isolated waveform events. By integrating nonlinear manifold learning, physiological trajectory reconstruction, unsupervised state discovery, probabilistic transition analysis, anomaly detection, toroidal cyclic embedding, orbital phase analysis, and holographic density rendering, the framework investigated whether physiological acoustic systems may exhibit measurable recurrent organization within latent topology.

Across the analyzed recordings, several recurrent structural properties emerged, including coherent latent manifold organization, recurrent physiological state regions, directional transition persistence, dynamically unusual anomaly structures, cyclic manifold recurrence behavior, orbital phase organization, and distributed occupation topology. Collectively, these observations suggest that physiological acoustic systems may exhibit substantially richer latent organizational structure than is immediately visible through conventional waveform inspection alone.

More broadly, the framework proposes a conceptual shift from static acoustic classification toward dynamical physiological trajectory modeling. Within this perspective, cardiac acoustic events become coordinates embedded within latent manifold geometry, transitions become structured pathways through physiological state-space, and recurrent acoustic evolution becomes distributed manifold traversal across dynamical topology.

Importantly, the framework remains exploratory and computational rather than clinically validated. The project does not claim diagnostic capability, pathological inference, physiological mechanism discovery, or medically validated latent topology. Similarly, concepts such as orbital recurrence, cyclic manifold organization, and partially stabilized physiological states should be interpreted as computational abstractions describing latent organizational behavior rather than direct biological mechanisms.

Instead, the broader contribution of the framework lies in proposing an alternative computational perspective in which physiological acoustic systems may be investigated as dynamically evolving structures embedded within latent geometric state-space. Within this emerging perspective, biomedical acoustic systems are explored not solely as diagnostic signals, but as trajectory-based dynamical organizations evolving throughout computational physiological topology.

10. References

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